

NAZMUL HUDA BADHON

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[Portfolio](#) | [LinkedIn](#) | [Google Scholar](#)

SUMMARY

Final-year Computer Science undergraduate (graduating Jan 2026), specializing in Deep Learning, Medical Imaging, and Medical AI. Author of 8+ peer-reviewed publications in Q1 and Q2 journals (Elsevier, Wiley), with additional works under review. Experienced in mentoring 30+ students through research training and academic support, and motivated to advance clinically impactful AI research through interdisciplinary collaboration, with a current focus on securing a Research Assistantship position.

EDUCATION

Daffodil International University, BSc in Computer Science and Engineering

• CGPA: 3.88/4.00

Spring 2022–Fall 2025

Dhaka, Bangladesh

Dongseo University, Exchange Program, Computer Engineering

• GPA: 3.94/4.50, Obtained Marks: 93.25% in total

Fall 2023

Busan, South Korea

RESEARCH INTERESTS

Deep Learning, Machine Learning, Medical Imaging, Medical AI.

RESEARCH PUBLICATIONS

Journal Publication

1. Salehin, I., **Badhon, N. H.**, Sajib, M. T. A., Moon, N. N. (2025). Secure Medical AI: Federated Learning and Large Language Models for Medical Intelligence. *Engineering Reports* (Wiley). **Submitted** [Q2].
2. Sajib, M. T. A., **Badhon, N. H.**, Salehin, I., Rifat, M. S. R., Ahmmed, F., Moon, N. N. (2025). A comparative deep learning methodology for plant insect image classification: assessment of CNN architectures and augmentation techniques. *MethodsX* (Elsevier). **Under Review** [Q2].
3. **Badhon, N. H.**, Salehin, I., Sajib, M. T. A., Rifat, M. H. S., Noman, S. M., Moon, N. N. (2025). TLCNN: Tabular data-based Lightweight Convolutional Neural Network for Electricity Energy Demand Prediction. *Global Energy Interconnection* (Elsevier). **Accepted** [Q1] (in press).
4. Salehin, I., **Badhon, N. H.**, Sajib, M. T. A., Moon, N. N. (2025). DeepUS-ReconSeg: A Multi-Angle Paired B-Mode Ultrasound Dataset for Medical Imaging Reconstruction and Segmentation. *Data in Brief* (Elsevier). **Published** [\[Link\]](#).
5. Salehin, I., Sajib, M. T. A., **Badhon, N. H.**, Rifat, M. H. S., Amin, N., Moon, N. N. (2025). Systematic Literature Review of LLM in Medical: Digital Health, Technology and Applications. *Engineering Reports* (Wiley). **Published** [Q2]. [\[Link\]](#)
6. Salehin, I., **Badhon, N. H.**, Sajib, M. T. A., Saifuzzaman, M., Moon, N. N. (2025). IQ-UltraRecon: Demodulated IQ Ultrasound Dataset of Human Hand and Arm Tissue for Deep Learning-Based Reconstruction. *Data in Brief* (Elsevier). **Published**. [\[Link\]](#)
7. Salehin, I., Khan, M. R., Habiba, U., **Badhon, N. H.**, Moon, N. N. (2024). BAU-Insectv2: An Agricultural Plant Insect Dataset for Deep Learning and Biomedical Image Analysis. *Data in Brief* (Elsevier). **Published**. [\[Link\]](#)

Book Chapter

8. **Badhon, N. H.**, Salehin, I., Sajib, M. T. A., Rifat, M. H. S., Moon, N. N. (2025). Predicting Peak Energy Demand Using Machine Learning Techniques for Efficient Electricity Supply Management. In *Engineering Applications of AI for Demand Forecasting* (Taylor & Francis). **Accepted** (in press).

PROJECTS

1. **Breast Cancer: Segmentation-Guided Hybrid Classification** [\[Link\]](#)
Built a two-stage U-Net and CNN-based hybrid model on BUSI datasets for lesion segmentation and classification, achieving strong performance on accuracy and ROC metrics.
2. **Lung–Colon Histopathology: Image Enhancement & Transfer Learning** [\[Link\]](#)
Applied preprocessing techniques on histopathology images and evaluated multiple transfer learning models, identifying the best-performing model by accuracy and F1-score.
3. **Image Preprocessing & Quality Analysis with Vision Transformer** [\[Link\]](#)
Designed preprocessing workflows for medical images and classified with ViT, achieving **87% testing accuracy** and improved transformer performance.

TECHNICAL SKILLS

Programming Languages: Python, C, Java

ML Frameworks: TensorFlow, PyTorch, Keras, Scikit-learn, OpenCV

Data Analysis & Visualization: NumPy, Pandas, Matplotlib, Seaborn

Tools: GitHub, Google Colab, LaTeX

TEACHING & MENTORSHIP EXPERIENCE

- Conducted 10+ research training sessions for 30+ student participants at the Multidisciplinary Action Lab, Daffodil International University.
- Assisted in delivering undergraduate academic content in Data Science, Database, and Big Data under faculty supervision.

AWARDS & LEADERSHIP

ACADEMIC

- Awarded a fully funded [Global Korea Scholarship \(GKS\)](#) for Exchange Study at Dongseo University by the Government of the Republic of Korea Fall 2023
South Korea
- Received a [Special Scholarship](#) for academic results from Daffodil International University Spring 2022–Fall 2025

RESEARCH

- Received a [Research Award](#) from the Division of Research, Daffodil International University 2024
- Served as a [peer reviewer](#) for *Digital Health* (SAGE Publishing) and received an official reviewer certificate 2025

LEADERSHIP

- Led a team of [20+ students](#) in International Camping at Dongseo University, South Korea 2023
- Completed the [Volunteer Internship Program](#) under the International Affairs, Daffodil International University 2024

REFERENCE

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